IPv4, IPv6 and Subnetting

1. Key Differences Between IPv4 and IPv6

IPv4	IPv6
Uses 32-bit addresses.	Uses 128-bit addresses.
Written in decimal, separated by dots. Example: 192.168.1.10	Written in hexadecimal, separated by colons. Example: 2001:db8::10
Provides about 4.3 billion total addresses.	Provides an extremely large address space (\(2^{128}\) addresses).
Uses subnet masks such as 255.255.255.0.	Uses prefix lengths such as /64.
Supports broadcast communication.	Does not use broadcast; it supports multicast and anycast.
NAT is commonly used to conserve public IPv4 addresses.	Provides a vast address space, reducing the need for address-conservation NAT.
ARP is used to resolve IPv4 addresses to MAC addresses on local networks.	Neighbor Discovery Protocol (NDP) performs similar functions.
IPsec support is optional.	IPsec support is part of the IPv6 standards, though its use is not mandatory in every deployment.

2. Assign Static IPv4 Addresses to Four Home Devices

Network: 192.168.10.0/24

Subnet mask: 255.255.255.0

Default gateway: 192.168.10.1

Device	Static IP Address	Subnet Mask	Default Gateway
Laptop	192.168.10.2	255.255.255.0	192.168.10.1
Smartphone	192.168.10.3	255.255.255.0	192.168.10.1
Smart TV	192.168.10.4	255.255.255.0	192.168.10.1
Printer	192.168.10.5	255.255.255.0	192.168.10.1

Explanation: Each device has a unique IP address in the same subnet. The default gateway is the router's address, which allows the devices to communicate with networks outside the home LAN.

Note: These addresses are an example plan. Make sure they do not conflict with the router's DHCP address pool or other statically assigned devices.

3. Subnetting Using a /27 CIDR Mask

Given network: 192.168.1.0/24

Required subnet mask: /27 = 255.255.255.224

Step 1: Calculate the number of borrowed bits

The original network is /24, and the new prefix is /27.

Borrowed bits:

27-24=3

Number of subnets:

2^3=8

Step 2: Calculate hosts per subnet

Host bits:

32-27=5

Total addresses per subnet:

2^5=32

Usable hosts per subnet:

32-2=30

Step 3: Calculate the block size

256-224=32

Therefore, the subnet network addresses increase by 32 in the last octet.

Subnet	Network Address	Usable Host IPs	Broadcast Address
1	192.168.1.0/27	192.168.1.1–192.168.1.30	192.168.1.31
2	192.168.1.32/27	192.168.1.33–192.168.1.62	192.168.1.63
3	192.168.1.64/27	192.168.1.65–192.168.1.94	192.168.1.95

Final answer: 8 subnets in total, with 30 usable host addresses per subnet, assuming the original network is 192.168.1.0/24.

4. Calculate Network Address, Broadcast Address and Host Range

Given:

- IP address: 10.0.5.17
- Subnet mask: 255.255.255.0

Step 1: Convert the subnet mask to CIDR

255.255.255.0 contains 24 network bits.

Therefore, the CIDR notation is /24.

Step 2: Calculate the network address

Perform a bitwise AND between the IP address and subnet mask.

IP address: 10.0.5.17

Subnet mask: 255.255.255.0

Network address: 10.0.5.0

The last octet becomes 0 because all host bits are zero.

Step 3: Calculate the broadcast address

Set all 8 host bits to 1. The last octet becomes 255.

Broadcast address: 10.0.5.255

Step 4: Calculate the valid host range

Number of host bits:

32-24=8

Usable host addresses:

2^8-2=256-2=254

Final answers:

Item	**Result**
CIDR	/24
Network address	10.0.5.0
Broadcast address	10.0.5.255
First valid host	10.0.5.1
Last valid host	10.0.5.254
Usable hosts	254

5. Verify the Answer Using an Online Subnet Calculator

Use the Subnet Calculator. It provides fields for the IP address, subnet mask, subnet ID and broadcast address.

subnet-calculator.com

Enter these values

IP Address

10.0.5.17

Subnet Mask

255.255.255.0

Expected results

- Subnet ID: 10.0.5.0
- Broadcast address: 10.0.5.255
- Host range: 10.0.5.1 – 10.0.5.254

Open Subnet Calculator

How to take your screenshot:

1. Open the calculator and enter the IP address and subnet mask.
2. Check that the subnet ID and broadcast address match the expected results above.
3. Press Win + Shift + S on Windows and select the calculator results.
4. Save or paste the screenshot into your assignment.